

Sourcing List — Automated Test Station

 **Print-ready PDF:** <docs/pdf/sourcing-list.pdf>

Shopping list for the breadboard prototype and first full build. Grouped by priority — Priority 1 blocks breadboarding and must be ordered before any bench work can start.

Legend:  on hand ·  ordered ·  to source

Approximate prices are USD, mid-2026 retail. Use as ballparks, not quotes.

Priority 1 — Order Today (blocks breadboarding)

These parts are required before you can do anything on the breadboard. Order all of them in one cart.

#	Item	Qty	Status	Approx \$	Notes
1	MAX31855 thermocouple amplifier breakout	3		~\$5 ea	Amazon search: " MAX31855 thermocouple amplifier breakout " — Adafruit #269 or compatible clone. Exposes VCC, GND, DO, CS, CLK on a 5-pin SIP header. Verify breakout has 3.3V regulator or is marked 3.3V-only — the MAX31855 is NOT 5V.
2	Solderless breadboard, 830-point, full-size	1		~\$8	Only if not already on hand. Amazon: " 830 point solderless breadboard ". A half-size (400-point) will be too tight for 3 modules + protection resistors — get full-size.
3	E-stop mushroom button, NC, latching, 22mm panel mount	1		~\$6	Amazon: " 22mm latching mushroom emergency stop button NC " — red head, normally-closed contacts. Must be latching (twist-to-release), not momentary. This is a safety-critical component — don't use the cheapest listing; pick one with 10A/250VAC contact rating.
4	Green momentary pushbutton, NC, 22mm panel mount	1		~\$3	Amazon: " 22mm momentary pushbutton green NC normally closed " — manual START button wired to LOGO! I4. NC type allows fail-safe wiring.
5	Red momentary pushbutton, NC, 22mm panel mount	1		~\$3	Amazon: " 22mm momentary pushbutton red NC normally closed " — manual STOP button wired to LOGO! I5.

Priority 1 subtotal: ~\$25–30

Priority 2 — Order Soon (needed for heater circuit)

Order these alongside Priority 1 or immediately after. The heater circuit cannot be tested or brought up without them.

#	Item	Qty	Status	Approx \$	Notes
6	Fotek SSR-40DA solid-state relay	1		~\$8-12	Amazon: " SSR-40DA solid state relay ". Spec: 3-32V DC control input, 24-480V AC load side, 40A, zero-crossing. At 150W/120VAC the load is 1.25A — well within rating. Must be genuine Fotek or a quality clone; cheap counterfeits have caused fires. Check seller reviews carefully.
7	150W 120VAC cartridge heater	1		~\$12-18	Amazon: " cartridge heater 150W 120V " or search Omega part " CIR-1012/120V ". Common size: 5/16" (8mm) diameter × 3" (76mm) long. Verify 120VAC specifically — 240VAC units exist and look identical.
8	KSD301 bimetallic thermostat disc, 120°C, NC	2		~\$1-2 ea	Amazon: " KSD301 thermostat 120C NC normally closed " — snap-disc type, mounts directly against heater body. 120°C trip point = hardware overtemp cutout wired to LOGO! I3. Buy 2: one installed, one spare.
9	Thermal cutoff fuse, 150°C, 10A, axial	5		~\$1-2 ea	DigiKey/Mouser: search " thermal cutoff fuse 150C 10A axial ". Or Amazon: " 152C thermal fuse 10A " — 152°C is a common standard rating close to 150°C. Spec: one-time non-resettable TCO type. Must be rated ≥10A. Clip directly to heater body — NOT in free air. Buy 5: 1 installed + 4 spares (a blown fuse means investigate; don't reuse).
10	Aluminium heatsink for SSR	1		~\$5-8	Amazon: " SSR heatsink aluminum solid state relay " — look for a dedicated SSR heatsink with pre-drilled mounting holes matching 40A SSR footprint (57.5mm × 44mm typical). Without a heatsink the SSR overheats at sustained duty. A DIN-rail mounted extrusion with fins ≥50 cm² face area also works.

Priority 2 subtotal: ~\$35-50

Priority 3 — Servo Vent Control

Order when you are ready to wire the vent servos. Not needed for initial heater bring-up.

#	Item	Qty	Status	Approx \$	Notes
11	Tower Pro MG90S metal-gear servo (or SG90 plastic)	2		~\$3–5 ea	Amazon: " MG90S metal gear servo " — metal gear recommended for repeated vent duty. SG90 plastic gear is fine for light loads. Standard 50Hz PWM, 500–2500µs pulse, 5V supply.
12	5V 2A BEC / DC-DC converter, 24V input	1		~\$5	Amazon: " 24V to 5V DC DC converter 2A step down " — powers servo rail from LOGO! Q2 24V relay output. Needs to handle 1A peak (2 servos at stall). Get a switching BEC, not a linear regulator (24→5V linear = 19V × 1A = 19W of heat = not good).

Priority 3 subtotal: ~\$12–20

Already Have (verify before ordering)

Check your bench stock before adding these to a cart.

#	Item	Qty	Status	Notes
13	K-type thermocouples with leads	3	✓	One per MAX31855 channel — TC1 ambient, TC2 DUT, TC3 heater. Yellow wire = positive (ANSI standard).
14	Dupont jumper wires (M-F, F-F assortment)	—	✓	For breadboard-to-RPi header connections.
15	Siemens LOGO! 12/24RCE PLC	1	✓	Already installed / on hand.
16	24V DIN rail PSU (Meanwell HDR-60-24 or equivalent)	1	✓	Powers LOGO! and relay coils.
17	WeMos D1 Mini / NodeMCU ESP8266	1	✓	Used for status-display sketch. NodeMCU (27mm row-spacing) will work on a breadboard; D1 Mini preferred for clean mounting.
18	Arduino resistor kit — 100Ω and 10kΩ	—	✓	Probably — confirm you have at least 3× 10kΩ (CS pull-ups) and 3× 100Ω (protection resistors). If stock is low, order a 1/4W resistor assortment on Amazon: "resistor assortment kit 1/4W 600 piece" .
19	100nF (0.1μF) ceramic capacitors	3+	✓	Probably — confirm at least 3× 100nF for MAX31855 VCC decoupling.
20	SSD1306 128×64 OLED display (I ² C)	1	✓	Check Arduino kit. If not on hand: Amazon "SSD1306 0.96 inch OLED I2C" ~\$4.
21	Raspberry Pi 5 4GB	1	?	Required for GPIO. Flask app runs on any host for dev, but SPI (MAX31855) and GPIO PWM (SSR, servos) need a Pi. See RPi note below.

Note on Raspberry Pi

Use a Raspberry Pi 5 4GB. The Pi 4 is heavily inflated on Amazon (\$125–150 for a kit). The Pi 5 is faster, more available, and at MSRP costs less.

Cheapest sourcing path (~\$78 total):

Item	Source	Price
RPi 5 4GB board	PiShop.us or Adafruit (\$60 MSRP)	\$60
27W USB-C PD power supply	Any 27W+ USB-C PD adapter — check your desk first. Official RPi PSU from PiShop ~\$12, or generic on Amazon ~\$8	\$8–12
32GB microSD A2-rated	Amazon: "Samsung EVO Select 32GB microSD"	\$8
Case	None needed for prototype — bare board on bench is fine	\$0

If buying a kit, the **Vilros RPi 5 Complete Starter Kit** (vilros.com) is \$134.99 on sale and includes board, 128GB SD, 27W PSU, aluminum case, and HDMI cables. Worth it if you want everything in one box — but \$55 more than buying parts individually.

One extra setup step for Pi 5: the standard `RPi.GPIO` library does not support the Pi 5's GPIO chip. Install the drop-in replacement before running the Flask app:

```
pip install rpi-lgpio --break-system-packages
```

This provides a fully compatible `RPi.GPIO` API on the Pi 5. No code changes required — just install it before `requirements.txt`.

SPI must be enabled: `sudo raspi-config` → Interface Options → SPI → Enable → reboot .

Total Cost Estimate

Priority bucket	Subtotal
Priority 1 — breadboard unblock	~\$25–30
Priority 2 — heater circuit	~\$35–50
Priority 3 — servos	~\$12–20
RPi 5 4GB + PSU + SD (if needed)	~\$78
Total to source	~\$150–178

Minimum spend to start breadboarding (Priority 1 only, RPi on hand): **~\$25–30**.